Huawei Access Node QoS Congestion Event SOP (MALL-ACC-03)

Node: MALL-ACC-03 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_1591

Description

The QoS congestion event on the Huawei Access Node (MALL-ACC-03) is a critical failure domain that occurs due to microbursts, interface tail drops, and QoS queue starvation. This phenomenon arises from the interaction between the node's scheduling algorithm, QoS configuration, and traffic patterns. The Huawei Access Node employs a complex scheduling mechanism to allocate bandwidth and prioritize traffic, taking into account factors such as traffic rates, burst sizes, and buffer sizes. However, under certain conditions, such as unexpected traffic spikes or misconfigured QoS policies, the node's scheduling algorithm may become overwhelmed, resulting in queue congestion, packet loss, and ultimately, the ALARM_1591 severity major alarm. Furthermore, the node's interface tail drops and QoS queue starvation can exacerbate the congestion event, leading to a vicious cycle of packet loss, buffer overflows, and scheduling algorithm instability. To mitigate this issue, it is essential to diagnose the root cause of the congestion event, identify the contributing factors, and implement targeted corrective actions to restore QoS and network stability.

Pre-requisite Checks

- check: Verify the redundant link status to ensure that the secondary link is not affected, description: Confirm that the secondary link is functioning correctly and that there are no issues with the redundant link configuration

- check: Check the environmental conditions, description: Verify that the node is operating within the recommended temperature and humidity ranges

- check: Gather relevant logs and performance data, description: Collect logs and performance data from the node, including interface counters, QoS statistics, and scheduling algorithm metrics

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Unexpected traffic spikes	Medium
Interface hardware issues	Low

Diagnostic Commands & Expected Outputs

```
Command: display scheduler profile, Expected Output: Scheduler profile configuration and statistics, Description: This command isolates the node's scheduling algorithm configuration, including the scheduling profile, priority policies, and scheduling metrics
```

```
Command: display queue statistics, Expected Output: Queue statistics, including queue depth, packet loss, and buffer utilization, Description: This command provides detailed statistics on the node's queues, including queue depth, packet loss, and buffer utilization, which can help identify the root cause of the congestion event

Command: display qos statistics, Expected Output: QoS statistics, including QoS policy hits, packet drops, and traffic classification metrics, Description: This command provides detailed statistics on the node's QoS policies, including QoS policy hits, packet drops, and traffic classification metrics, which can help identify the contributing factors to the congestion event
```

Resolution Steps

Step 1: Step: Reconfigure the QoS policies, Command: config qos policy , Description: Reconfigure the QoS policies to prioritize traffic correctly and ensure that the scheduling algorithm is functioning correctly

Step 2: Step: Adjust the scheduling algorithm parameters, Command: config scheduler profile , Description: Adjust the scheduling algorithm parameters to optimize the node's scheduling mechanism and prevent congestion events

Step 3: Step: Disable the redundant link, Command: undo interface , Description: Disable the redundant link to isolate the issue and prevent further packet loss and congestion events

Post-Check Verification

```
Command: display scheduler profile, Expected Output: Scheduler profile configuration and statistics, Description: Verify that the scheduling algorithm configuration has been updated correctly and that the scheduling metrics are within the recommended range

Command: display queue statistics, Expected Output: Queue statistics, including queue depth, packet loss, and buffer utilization, Description: Verify that the queue statistics have improved and that packet loss and buffer utilization are within the recommended range

Command: display qos statistics, Expected Output: QoS statistics, including QoS policy hits, packet drops, and traffic classification metrics, Description: Verify that the QoS statistics have improved and that QoS policy hits, packet drops, and traffic classification metrics are within the recommended range
```

Roll-back Procedures

- Step: Revert the QoS policies, Command: undo config qos policy , Description: Revert the QoS policies to the previous configuration in case the new configuration causes further issues

- Step: Revert the scheduling algorithm parameters, Command: undo config scheduler profile , Description: Revert the scheduling algorithm parameters to the previous configuration in case the new configuration causes further issues

- Step: Re-enable the redundant link, Command: undo undo interface , Description: Re-enable the redundant link to restore redundancy and prevent further packet loss and congestion events